GGG-S-121F
January 20, 1992
SUPERSEDING
GGG-S-121E
June 10, 1974
(see section 6)

FEDERAL SPECIFICATION

SCREWDRIVER AND SCREW STARTER, HAND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers screwdrivers intended for manual operation in driving or removing screws with single or cross recesses. Screw starters are used to start or remove screws primarily in space-restricted areas. The screwdrivers and screw starters shall be of the types normally used by cabinet makers, laborers, sheet metal workers, production workers, mechanics, and electrical repairmen.

1.2 Classification.

1.2.1 Types, classes, designs, and styles. Screwdrivers and screw starters shall be of the following types, classes, designs, and styles as specified (see 6.1). (Note: See 6.2 for deleted items.)

Type I - Flat tip
  Class 4 - Heavy duty, flared sides
    Style 2 - Steel handle with wood inserts
  Class 6 - Flared sides with screw starter, external gripping
  Class 7 - Pocket, removable blade

Type III - Screw starters, hand

Type IV - Offset
  Class 1 - Double offset, single flat tip each end
  Class 2 - Four-way offset, double flat tip each end
  Class 3 - Phillips, cross tip each end
    Style 1 - Double offset
    Style 2 - Single offset

Type V - Ratchet
  Class 1 - Common, straight

Type VI - Cross tip, straight
  Class 2 - Reed and Prince
  Class 3 - Phillips, with screw starter, external gripping, with cushion grip

Type VII - Screwdriver, spark testing

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

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Federal Standards:

FED-STD-376	- Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)

B18.6.3	- Screw, Machine, and Screw Nuts, Machine, Slotted and Recessed Head
B46.1	- Surface Texture (Surface Roughness, Waviness and Lay)

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society of Mechanical Engineers (ASME)

ANSI/ASME B107.4	- Driving and Spindle Ends for Portable Hand, Air, and Electric Tools
ANSI/ASME B107.15	- Flat Tip and Phillips Screwdrivers

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

American Society for Testing and Materials (ASTM):

D 2240	- Standard Method of Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer
E 18	- Metallic Materials Rockwell Hardness and Rockwell Superficial Hardness of

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are illustrative and not restrictive provided the tools are otherwise in accordance with this specification.

3.2 Marking. Screwdrivers shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, molded, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item. Screwdrivers shall not be marked with wording such as "shockproof" which may imply to the user that the handle provides complete electrical protection.

3.3 Test loads. When tested as specified in 4.4.2 through 4.4.3, the tools shall withstand the test loads of the applicable tables without breakage or permanent visible distortion.

3.4 Blades. Blades shall be of steel, properly heat treated to withstand the tests specified herein. All four sides, and the end of flat tip blades, shall be symmetrical. The tip thickness shall be uniform across its entire width within 0.003 inch. The tang end of the blades shall be held in the handle securely, and in line with the handle within one degree angular variation of the handle and blade axis. When specified (see 6.1), blades shall be covered with an insulating material for the purpose of preventing short circuiting of electrical apparatus while adjustments or maintenance is being performed. Insulation covering shall meet the dielectrical strength test in 4.4.10.

3.4.1 Finish. The blades shall have a smooth, bright finish that will produce a 125 microinch roughness height rating or better (as determined in accordance with ANSI B46.1). The blades shall be treated in a manner to prevent rust or corrosion. Plating, if any, shall be adherent. Cadmium plating will not be acceptable.

3.4.2 Hardness. Unless otherwise specified herein, the tip portion of the screwdrivers, or the entire blade if it is the manufacturer's standard practice, shall be hardened to not less than 50 nor more than 58 on the Rockwell C scale when tested as specified in 4.4.1.

3.5 Plastic handles. The handles shall be made of a material capable of withstanding the applicable test requirements of 4.4.5, and shall be suitably finished to provide a comfortable grip. Cross-sections shall be essentially square, hexagonal, triangular, round, or octagonal. Square, hexagonal, and triangular handles shall be rounded on the edges to a minimum radius equal to the distance across the flats divided by four. Round and octagonal handles shall be fluted with four to six major flutes for a length at least 65 percent of the handle length. The size of the flute and the major periphery shall make an angle of not less than 90°, and the juncture shall be broken to 1/64 radius minimum. The handle butt shall be well rounded to a minimum radius to 1/8 of the major cross-sectional dimension and the run out of the radius with the major periphery shall not exceed an angle of 15°. The handle volume shall be at least 53 percent of the major cross-sectional dimension times the length for standard handles, and 50 percent for stubby handles.

3.6 Handle with 3/8 inch female drive insert. When specified (see 6.1), handles shall have a metal 3/8 inch square female drive insert of a design suitable for accepting a corresponding 3/8 inch square male drive tang in accordance with ANSI B107.4, table 9. The insert shall be flush with the butt end of the screwdriver handle. If the insert is made of ferrous material, it shall be treated to prevent corrosion. The insert shall meet the test requirements in 4.4.8. When handles with inserts are furnished, the diameter of the handle shall not be less than 1-1/32 inch. One or more faces of the female drive opening shall be drilled or recessed so that it will engage the spring-loaded steel ball, or steel plunger on the corresponding male drive.

3.7 Cushion grip. When specified (see 6.1), handles shall be furnished with cushion grips. The grips shall be of rubber or vinyl capable of meeting the test requirements of 4.4.5. The length shall be at least 60 percent of the handle length, and there shall be not detectable slippage between the handle and the cushion grip under a normal usage test. The cushion grip shall not project past the butt end of the handle, and both ends of the cushion grip shall blend smoothly with the handle. The cushion grip shall be ribbed or fluted on the exterior surface to prevent slippage of the hand. The cushioned handles shall meet the dimensional requirements for corresponding conventional type handles for each type, class, and style (see applicable tables) except that an additional 7/8 inch plus tolerance will be permitted on the handle length, and 3/16 inch plus tolerance will be permitted on the diameter of the handle. The fluting requirements for handles do not apply when cushion grips are supplied.

3.8 Type I, flat tip. The blades of flat tip screwdrivers shall be in accordance with 3.4, 3.4.1, and 3.4.2. The included angle between bearing surfaces shall be 10° maximum. The extreme tip width shall be perpendicular to the blade axis to within 2°, and the tip thickness to within 6°. The tip thickness shall be uniform to within 0.003 inch.

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3.8.1 Type I, class 4, style 2, heavy duty, flared sides. The blade and handle frame of the type I, class 4, style 2 screwdriver shall be forged in one piece from high-grade steel. A hardwood insert shall be secured to each side of the handle frame with not less than two rivets. The rivet heads and outer surface of the inserts shall be finished flush with the outer surface of the handle frame. The cross-sectional shape of the handle shall be approximately elliptical, smoothly and uniformly finished with the greater diameter of the ellipse being in alignment with the narrow edges of the flat tip of the blade. The inserts shall be protected at both ends by the metal portion of the handle. The wood shall be clear, straight-grained, white ash, beech, birch, hickory, hard maple, or pecan. Moisture content shall not exceed 12 percent. Handles shall be free of defects except that mineral streaks and sound, tight knots of not more than 1/8 inch diameter will be allowed. The screwdriver shall conform to table I and be similar to figure 1.

TABLE I. Type I, class 4, style 2, heavy duty, flared sides

Blade			Handle		Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XV)
Exposed length ± 1/4	Tip width at end + 1/16 - 1/32	Thickness of tip ± 0.004	Width at largest section + 1/4 - 1/16	Length (Min.)			
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
4	5/16	0.042	1-1/8	4	80	700	G
6	3/8	0.050	1-3/8	4	140	1,000	H
8	7/16	0.058	1-1/2	5	200	1,200	I
10	7/16	0.058	1-1/2	5	200	1,200	I
5 1/	1/2	0.070	1-3/8	4	250	1,400	J

1/ Extra heavy duty. Furnished with square blade, and shot-blast finish.

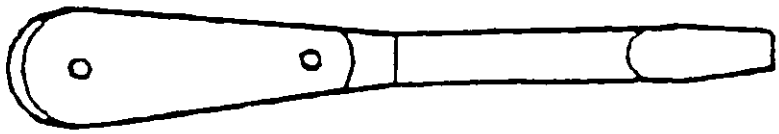


Figure 1. Type I, class 4, style 2, heavy duty, flared sides.

3.8.2 Type I, class 6, flared sides with screw starter, external gripping. Type I, class 6 screwdrivers shall be suitable for holding screws when starting or removing screws from restricted places, and shall be suitable for normal screwdriver work. The screwdriver shall consist essentially of a blade, screw-gripping mechanism, and a handle. The screw gripper shall be of the spring-action type capable of holding any size screw for which the point thickness of the blade is suited, and passing the tests of 4.4.7. The gripper shall slide the full length of the blade, and shall have sufficient friction to remain in any position on the shank. The handle shall conform to 3.5. The screwdriver shall conform to table II, and be similar to figure 2.

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TABLE II. Type I, class 6, flared sides with screw starter, external gripping

Blade			Handle		Torsional moment (Min.)	Bending moment (Min.)	Test block (see table XV)
Exposed length ± 1/4	Tip width at end + 1/32 - 1/64	Thickness of tip at end ± 0.004	Diameter or distance across corners (Min.)	Length (Min.)			
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	Inch-Pounds	
2	5/32	0.025	9/16	2	6	60	C
4	5/32	0.025	9/16	2-5/8	12	60	C
6	5/32	0.025	9/16	2-5/8	12	60	C
10	5/32	0.025	9/16	2-5/8	12	60	C
4	7/32	0.032	15/16	3-1/4	30	175	E
8	5/16	0.042	1-1/16	3-7/8	80	500	G

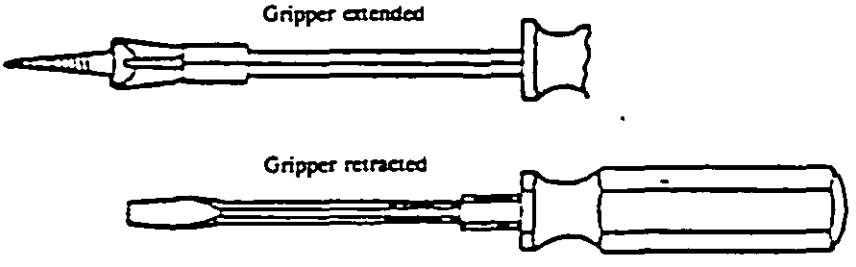


Figure 2. Type I, class 6, flared sides with screw starter, external gripping.

3.8.3 Type I, class 7, pocket, removable blade. Type I, class 7 screwdrivers shall be of steel and shall consist essentially of a handle, blade, and a chucking device. The body of the handle shall be of a round cross-sectional shape, and shall be neatly knurled. External surfaces shall be protected by nickel plating or other suitable and equally effective means. The upper portion of the handle shall be provided with a knob at least 1/8 inch larger than the diameter of the handle. The handle shall be provided with a chuck so designed that the shank of the blade may be readily locked in the handle. The design shall also permit the blade to be inverted and inserted in the handle when not in use. Additional blades shall be furnished when specified (see 6.1). The screwdrivers shall conform to table III, and be similar to figure 3.

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TABLE III. Type I, class 7, pocket, removable blade

Size number	Blade			Body Diameter ± 1/32	Closed length (Min.)
	Tip width at end + 0.015 - 0.010	Thickness of tip at end + 0.003 - 0.002	Length under chuck ± 1/4		
	Inch	Inch	Inches	Inch	Inches
1	0.090	0.010	1	1/4	3
2	0.156	0.014	2	3/8	4

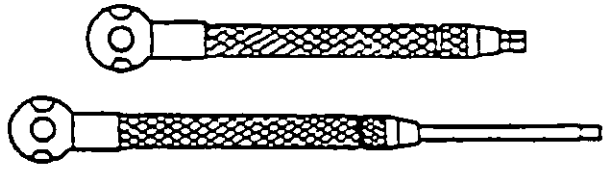


Figure 3. Type I, class 7, pocket, removable blade.

3.9 Type III, screw starters, hand. Type III screw starters shall be for use on straight-slotted screws, and shall consist essentially of a screw-slot gripping mechanism and a handle.

3.9.1 Screw-slot gripping mechanism. The screw-slot gripping mechanism shall grip the screw internally within the slot of the screw head by a wedge action between the blade and the screw slot, and shall be suitable for starting or removing screws from restricted places. The tip and the part of the gripper which engages the screw slot shall have a hardness of not less than 45 and not more than 58 on the Rockwell C scale.

3.9.2 Handle. The handle shall be the portion of the screw starter that is held in the hand, and shall be made of either plastic or metal of a suitable size. Screwdriver type handles shall be in accordance with 3.5. Metal handles shall be knurled.

3.9.3 Dimensions. Type III screw starters shall conform to the dimensions in table IV for the size specified (see 6.1).

TABLE IV. Type III, screw starter, hand

Tip width at end + 1/16 - 1/64	Thickness of tip at end ± 0.004	Overall length (Min.)
Inch	Inch	Inches
1/8	0.017	6-3/4
3/16	0.030	3-5/8
9/32	0.032	5-1/8
1/4	0.037	7-1/4
1/4	0.037	9-1/4
9/32	0.032	9-1/4
3/16	0.030	11-1/4

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3.10 Type IV, offset.

3.10.1 Class 1, double offset, flat tip each end. Class 1 screwdrivers shall be of one piece steel construction. The flat tips shall be at right angles to each other, and conform to 3.4. The cross-sectional shape of the body shall be octagon, hexagon, or square. When hexagon or square-shaped stock is used, the edges along the entire body length shall be well rounded to provide a comfortable grip. In lieu of octagon, hexagon, or square-shaped body, round will be acceptable, provided it is well knurled or has a coarse shot-blast finish for firm gripping. The screwdrivers shall conform to table V and be similar to figure 4.

TABLE V. Type IV, class 1, double offset

Tip width at end + 1/32 - 1/64	Offset ± 1/4	Thickness of tip at end ± 0.004	Length ± 1/2	Diameter or distance across flats of body ± 1/32	Torsional moment (Min.)
Inch	Inch	Inch	Inches	Inch	Inch-Pounds
5/32	1/2	0.025	3	5/32	12
1/4	1/2	0.037	4-1/4	7/32	40
5/16	5/8	0.042	5	9/32	80
3/8	3/4	0.050	6	11/32	140
1/2	7/8	0.070	6	13/32	250
3/4	1	0.130	7-1/2	1/2	400

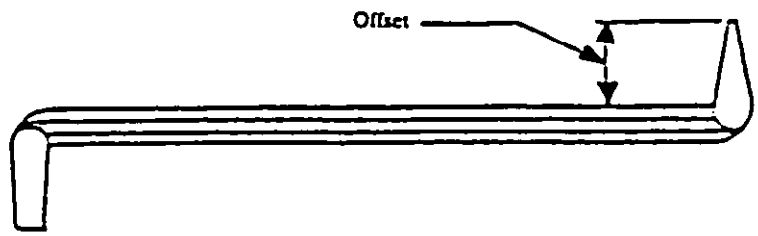


Figure 4. Type IV, class 1, double offset.

3.10.2 Type IV, class 2, four-way offset, flat tips each end. Class 2 screwdrivers shall be of steel, and may of one piece construction or have blades securely fastened into the ends. The tips shall be arranged in such a manner that eight positions in 360° are available to engage screw slots. There shall be four blades, two at each end, protruding in opposite directions with one blade being in line with the longitudinal axis of the body and the other blade at right angles to the axis. The two blades at the other end of the body shall be parallel to each other and set at 45° to the axis of the body. The axis of the blades at both ends shall be parallel. The tips and hardness shall be in accordance with the applicable requirements of 3.4 and 3.4.2 and the blade tips shall be flared. The cross-sectional shape of the body shall conform to 3.10.1. The screwdriver shall conform to table VI, and be similar to figure 5.

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TABLE VI. Type IV, class 2, four-way offset

Tip width at end ± 1/64	Offset + 1/4 - 1/16	Thickness of tip at end ± 0.004	Length + 3/8 - 1/4	Diameter or distance across flats of body + 3/16 - 1/16	Torsional moment (Min.)
Inch	Inch	Inch	Inches	Inch	Inch-Pounds
1/4	5/16	0.037	5-1/4	5/16	40

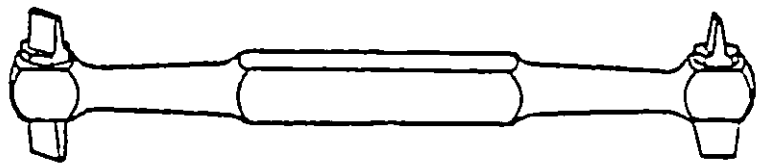


Figure 5. Type IV, class 2, four-way offset.

3.10.3 Type IV, class 3, style 1, double offset, Phillips cross tip each end. The class 3 screwdriver shall have Phillips type points formed on opposite sides of the tool. The screwdrivers shall be of one piece steel construction and be capable of withstanding the tests specified in 4.4.1 through 4.4.2.1. Each cross tip blade shall show a hardness of not less than 52 nor more than 60 on the Rockwell C scale. The reading shall be taken as near the tip end as practicable.

3.10.3.1 Phillips screwdrivers. Phillips screwdrivers and bits shall conform to the dimensions shown on Phillips Drawing PD-9 or PD-18, as applicable. The screwdrivers shall conform to table VII, and be similar to figure 6.

TABLE VII. Type IV, class 3, style 1 screwdriver, double offset, Phillips cross tip each end

Point size		Length of each offset		Overall length ± 1/4	Diameter of shank (Max.)		Torsional moment	
Small end	Large end	Small end ± 3/16	Large end ± 3/16		Small end	Large end	Small end (Min.)	Large end (Min.)
Number	Number	Inches	Inches	Inches	Inch	Inch	Inch-Pounds	Inch-Pounds
1	2	3/4	7/8	4-3/4	0.197	0.260	25	60
2	3	7/8	1-1/16	5-1/2	0.260	0.322	60	150
3	4	1-1/16	1-5/16	6	0.322	0.385	150	200

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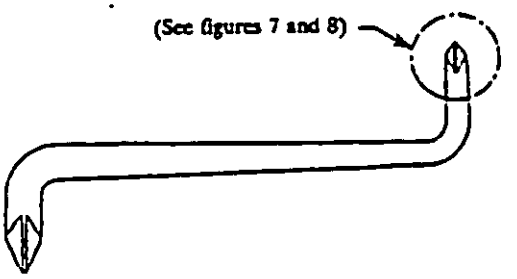
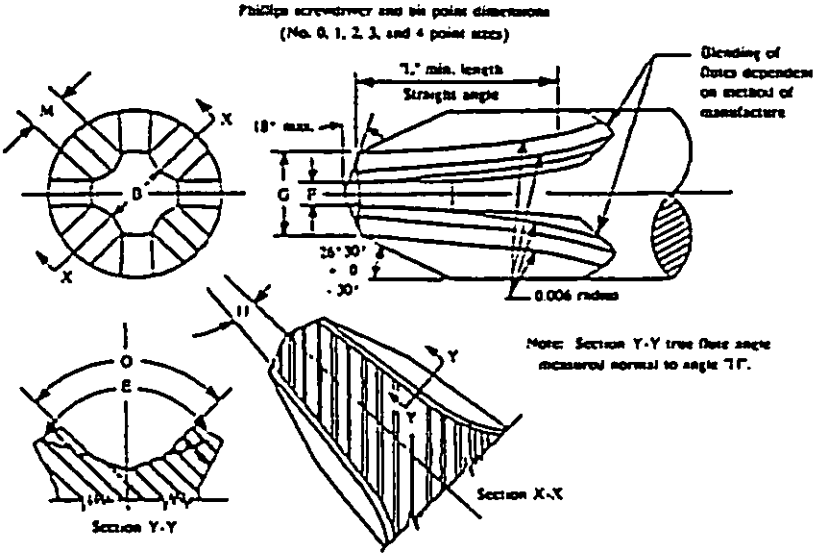


Figure 6. Type IV, class 3, style 1, double offset, Phillips cross tip.

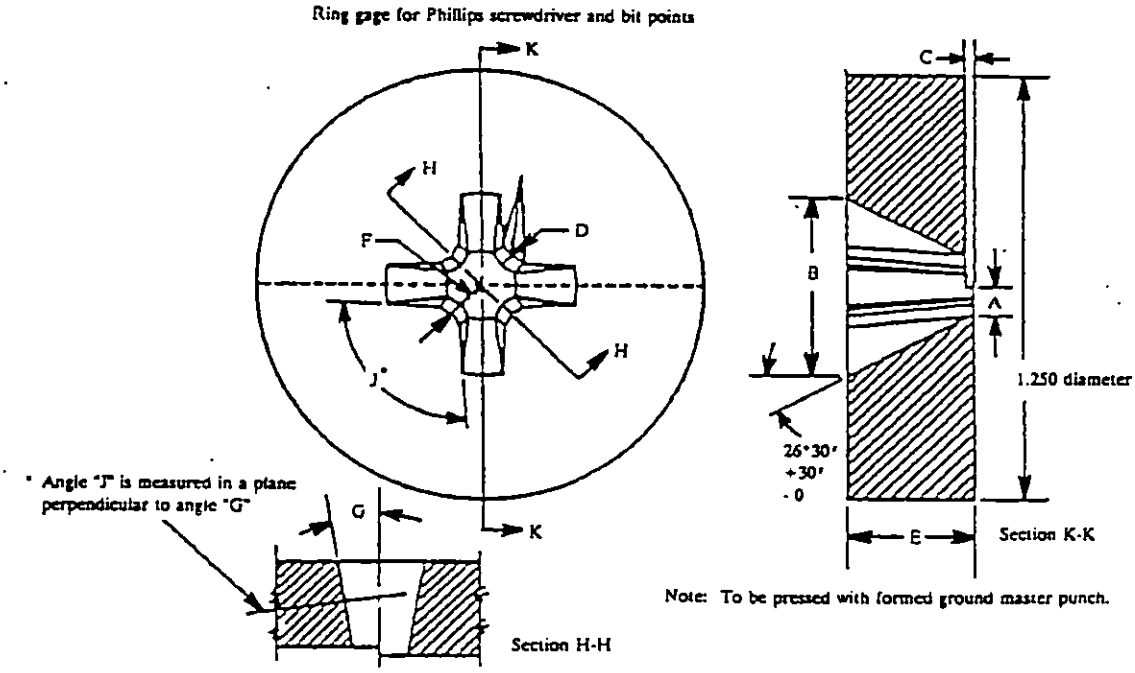


Point size	B ± 0.001	E + 30′ - 0	G ± 0.001	H + 0 - 30′	F	O + 30′ - 0	M ± 0.001	L
Number	Inch	Degree	Inch	Degree	Inch	Degree	Inch	Inch
0	0.0230	See note	0.032	7°	0.012 0.014	92	0.0078	1/8
1	0.0394	138	0.050	7°	0.016 0.020	92	0.0202	1/8
2	0.0606	140	0.090	5°45′	0.023 0.028	92	0.0434	3/16
3	0.0983	146	0.150	5°45′	0.027 0.033	92	0.0826	9/32
4	0.1407	153	0.200	7°	0.040 0.045	92	0.1078	11/32

Note: For No. 0 size E equals 0.004 radius approximate.

Figure 7. Phillips screwdriver point dimensions and requirements.

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Point size	A ± 0.0002	B (Ref)	C ± 0.001	D (Ref)	E + 0 - 1/64	F ± 0.0002	G + 15' - 0	J
Number	Inch	Inch	Inch	Inch	Inch	Inch	Degree	Degree
0	0.0140	0.1230	0.010	0.0268	3/32	0.0114	7°	92
1	0.0255	0.1785	0.010	0.0634	1/8	0.0194	7°	92
2	0.0455	0.2800	0.010	0.0971	3/16	0.0303	5°45'	92
3	0.0755	0.4348	0.010	0.1538	9/32	0.0496	5°45'	92
4	0.1005	0.5471	0.010	0.2083	11/32	0.0711	7°	92

Figure 8. Ring gages for inspecting Phillips screwdriver and bit points (see figure 6).

3.10.4 Type IV, class 3, style 2, single offset. The class 3, style 2, screwdrivers shall have only one end offset and the points on each end shall be the same size. Style 2 screwdrivers shall conform to the requirements in table VIII, and be similar to figure 9.

TABLE VIII. Type IV, class 3, style 2 screwdriver, single offset, Phillips cross tip each end

Point size	Length of offset ± 3/16	Overall length ± 1/4	Torsional moment (Min.)
Number	Inches	Inches	Inch-Pounds
2	1-1/8	4	60

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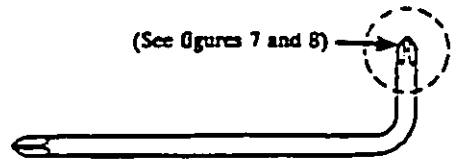


Figure 9. Type IV, class 3, style 2 screwdriver, single offset, Phillips cross tip each end.

3.11 Type V, ratchet.

3.11.1 Class 1, common, straight. The class 1 screwdrivers shall be those known commercially as ratchet screwdrivers. They shall consist essentially of a blade, handle, ratchet mechanism, and a shifter. The blade shall conform to 3.4, 3.4.1, and 3.4.2. The gripping portion of the handle shall be smoothly finished and suitably shaped to provide a good handgrip. The ratchet mechanism shall be of a design to permit the screwdriver to operate as a right- or left-handed ratchet screwdriver, or as a rigid tool. The screwdriver shall conform to table IX, and be similar to figure 10.

TABLE IX. Type V, class 1, common, straight

Blade			Handle		Torsional moment (Min.)	Test block (see table XV)
Exposed length ± 1/4	Tip width at end + 1/32 - 1/64	Thickness of tip at end ± 0.004	Diameter or distance across corners (Min.)	Length (Min.)		
Inches	Inch	Inch	Inches	Inches	Inch-Pounds	
3	5/16	0.042	1-1/4	3-1/2	60	G
6	5/16	0.042	1-1/4	4	60	G

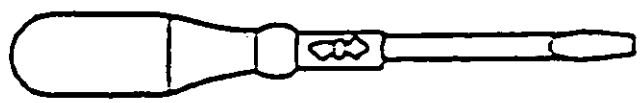


Figure 10. Type V, class 1, common, straight.

3.12 Type VI cross tip straight.

3.12.1 Class 2, Reed and Prince cross-point bits. When specified (see 6.1), bits with Reed and Prince type points shall be furnished. They shall be of one piece steel construction, and capable of withstanding the tests specified in 4.4.1 and 4.4.2.1. Each cross tip blade shall show a hardness as specified in 3.10.3. The bits shall be capable of tightening and loosening screws having a Reed and Prince recess in accordance with ANSI B18.6.3. The chuck end of the bit shall conform to table X, and be similar to figure 11. When specified (see 6.1), bits shall be supplied separately. In addition to the marking requirements of 3.2, the screwdriver bits shall be permanently marked with the name "Reed and Prince". The screwdriver tip shall conform to the dimensions shown on figure 12. The screwdriver shall have plastic handles in accordance with 3.5, conform to the dimensions specified in table XI, and be similar to figure 13.

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TABLE X. Chuck end dimensions

Nominal diameter for chuck end of bits	A ± 0.010	B + 0.010 - 0.020	C + 0.015 - 0.020	D ± 0.009	E ± 0.015
Inch	Inch	Inch	Inch	Inch	Inch
7/32	0.555	0.270	0.115	0.089	0.075
9/32	0.650	0.330	0.115	0.119	0.085
5/16	0.650	0.355	0.115	0.116 - 0.142	0.085

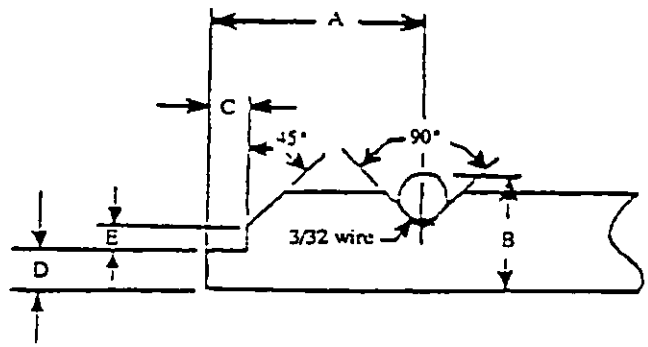
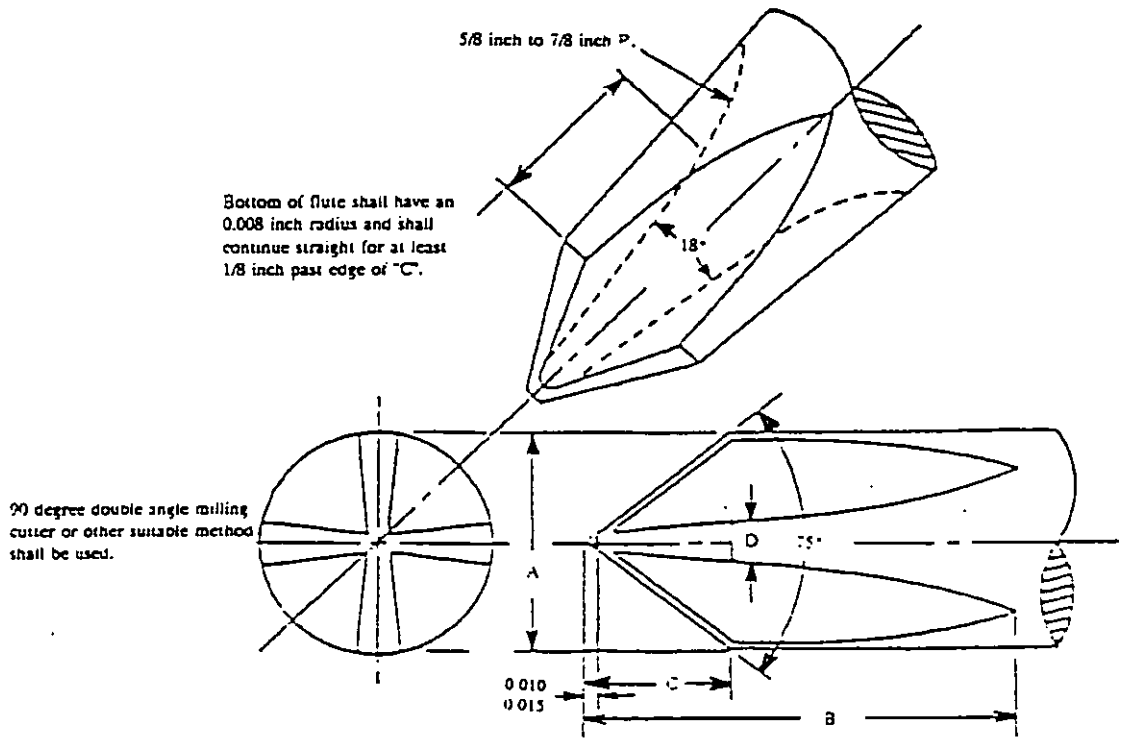


Figure 11. Chuck end of bits and adapter.



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Blade size			
A	B	C	D
Inch	Inch	Inch	Inch
0.187	1/2	0.122	0.041
0.250	9/16	0.163	0.050
0.312	5/8	0.204	0.060
0.375	11/16	0.244	0.069

Figure 12. Reed and Prince screwdriver-point dimensions and requirements.

TABLE XI. Type VI, class 2 screwdriver, cross tip, straight, Reed and Prince

Blade		Handle		Torsional moment (Min.)
Exposed Length ± 1/4	Diameter ± 0.010	Diameter (Min.)	Length (Min.)	
Inches	Inch	Inches	Inches	Inch-Pounds
3	3/16	15/16	3-1/4	25
10	3/16	1	3-1/2	25
4	1/4	1	3-1/2	40
6	5/16	1-1/8	3-1/2	90
15	5/16	1-1/8	4-5/8	90
8	3/8	1-1/8	4-5/8	150
1 1/	3/16	13/16	1-7/8	15
1-1/2 1/	1/4	13/16	1-7/8	20

1/ Close quarter screwdrivers.

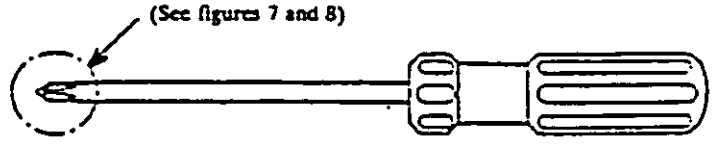


Figure 13. Type VI, class 2 screwdriver, cross tip, straight, Reed and Prince.

3.12.2 Class 3, Phillips, with screw starter, external gripping, plastic handle, with cushion grip. The type VI, class 3 screwdriver shall have Phillips type points as specified in 3.10.3. The blade shall be round in cross section, formed from high grade steel, hardened and marked to conform to the requirements of 3.10.3, and be capable of withstanding the test specified in 4.4.2.1. The screwdrivers shall have

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plastic handles in accordance with 3.5, conform to table XII, and be similar to figure 14, as specified (see 6.1). The screwdriver handle shall be provided with a cushion grip in accordance with 3.7, and the blade and shank shall be provided with a gripping mechanism in accordance with 3.8.2.

TABLE XII. Type VI, class 3 screwdriver, cross tip, straight, Phillips

Blade			Handle		Torsional moment (Min.)
Exposed length ± 1/4	Point size	Diameter ± 0.010	Diameter (Min.)	Length (Min.)	
Inch	Number	Inch	Inches	Inches	Inch-Pounds
3	1	3/16	13/16	3-1/4	25
4	2	1/4	7/8	3-1/2	40

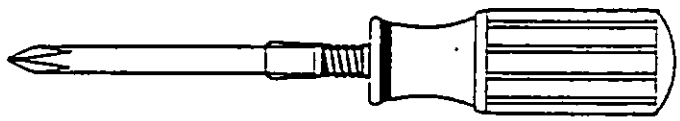


Figure 14. Type VI, class 3 Phillips, with screw starter, external gripping, plastic handle, with cushion grip.

3.13 Type VII, spark testing. Type VII screwdrivers shall be suitable for testing spark plugs in automobile engines and other gasoline engines. They shall consist essentially of a blade and plastic handle with a neon tube. The blade shall withstand the hardness and torsional moment tests of 4.4.1 and 4.4.2. The handle shall not be subjected to an impact test. A pocket clip shall be furnished on the 2-1/2 inch blade length screwdriver. The clip shall be of corrosion-resistant metal, or treated to resist corrosion, and shall be either a slip-on type or permanently secured to the screwdriver handle. The movable end of the pocket clip shall have a rounded section which shall contact the handle within 0.006 inch. The clip shall have sufficient spring action to return to its normal position after a 3/16 inch diameter rod is passed between the handle and the clip end. The screwdriver shall conform to table XIII and be similar to figure 15.

TABLE XIII. Type VII screwdrivers, spark testing

Blade dimensions			Handle dimensions		Torsional moment (Min.)
Exposed length ± 1/2	Diameter and width of tip at end + 0.065 - 0.005	Thickness of tip at end ± 0.004	Diameter (Min.)	Length (Min.)	
Inches	Inch	Inch	Inch	Inches	Inch-Pounds
2-1/2	1/8	0.020	7/16	2	6

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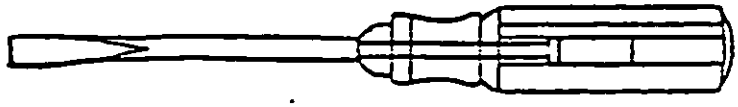


Figure 15. Type VII screwdriver, spark testing.

3.13.1 Blade. The blade for type VII screwdrivers shall conform to 3.4, 3.4.1, and 3.4.2. The cross-sectional shape shall be round and the tip shall be either straight or flared.

3.13.2 Handle. The handle shall conform to the applicable requirements of 3.5. The transparency of the handle shall permit the light flashes of the neon tube within the handle to be plainly visible.

3.14 Workmanship. The screwdrivers shall conform to the requirements in this specification. They shall be free from rust and defects which may impair their serviceability, durability, or appearance.

3.15 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. The end items shall be examined for the defects listed in table XIV. The lot size shall be expressed in terms of screwdrivers. The sample unit shall be one screwdriver. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

TABLE XIV. End item visual examination

Examine	Defect
Construction and Workmanship	Not as specified
	Sides, and end of flat tip, not symmetrical
	Blade tip not as specified
	Tang not secure

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TABLE XIV. End item visual examination (cont'd)

Examine	Defect
Construction and Workmanship (cont'd)	Components missing (as applicable) Insert missing, when specified Cushion grip not properly fitted Gripper does not hold screw (type I, class 6; type III; type VI, class 3) Blade does not lock in handle or cannot be inserted in handle (type I, class 7)
Material	Not as specified Blades not steel; not properly heat-treated Hardwood inserts improper type of wood (type I, class 4) Plating not suitable (type I, class 7)
Finish	Not as specified Blade not bright, smooth Blade not corrosion-resistant Plating not adherent Handle grip not comfortable Knurling missing (type I, class 7)
Marking	Missing; incomplete; not legible; not permanent

4.3.2 Dimensional examination. The end items shall be examined for any nonconformance to dimensional requirements. The lot size shall be expressed in terms of screwdrivers. The sample unit shall be one screwdriver. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.3.2.1 Conformance to dimensional requirements. Conformance requirements may be verified either by actual measurement of the tip configuration, or by gaging with a ring gage conforming to Phillips Drawing PG-2 or PG-12, as applicable, which has been prechecked with a Phillips master gage conforming to Phillips PG-3 or PG-13, as applicable. The gage shall fit on the master gage firmly (without wobble) and shall be representative of the fit required of the driver checked in the ring gage. In addition, the driver shall fit the ring gage so that the "G" dimension (on the Phillips drawing) of the tip falls within the step on the face of the ring gage. The lot size shall be expressed in units of screwdrivers. The sample unit shall be one screwdriver. The inspection levels shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.4 Test methods. The end items shall be tested for the characteristics specified in 4.4.1 through 4.4.10, as applicable. The lot size shall be expressed in units of screwdrivers. The sample unit shall be one screwdriver. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.1 Hardness. The Rockwell hardness tests shall be conducted in accordance with ASTM E 18. Care shall be exercised to assure that the flat side of the blade or bit under test is normally at right angles to the axis of the penetrator of the testing machine.

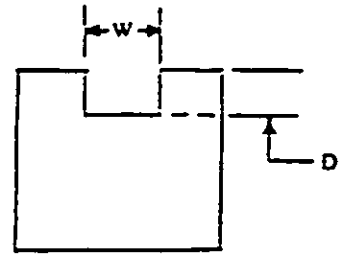
4.4.2 Torsional moment test. The sample screwdriver shall be placed in an oven with the thermostat set for 125°F. The screwdriver shall remain in the oven for one hour. Then it shall be removed and within one minute the torsional moment test shall be conducted in the following manner: The blade end of each sample item under test shall be located in the applicable test block shown in table XV. The stress shall be applied by a force acting at or near the middle of the natural grip of the handle, the force to act substantially at right angles to the axis of the screwdriver or screw starter. The stress shall be applied with the blade end held securely in the fixed slot while force is applied to an arm, bar, or clamp secured to the handle; or the stress may be applied by a force applied to the handle, under which the screwdriver or screw starter will be required to lift a dead weight attached to a lever arm, or in an equally satisfactory manner which will subject the tool to a known torsional stress from the handle end while the blade end is located in the test block. The force shall be applied

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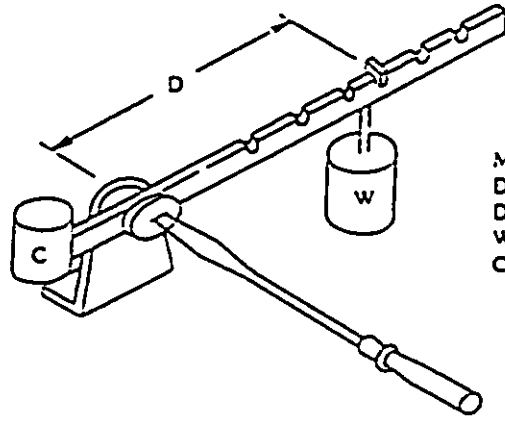
by means of a dead weight, through a spring-balance, or other suitable load-measuring device. The torsional moment shall be applied in right- and left-hand directions alternately for type V, class 1 screwdrivers. The torsional moment shall be computed by multiplying the force in pounds by the arm in inches (15-pound load applied 10 inches from axis of tool = 150 inch-pounds, and so forth). The arm is the distance from the axis of the tool to the direction of the force at right angles to the latter. Figure 16 illustrates a device and the method employed in conducting torsional moment tests on screwdrivers or screw starters. It is permissible to support the blade at or near the junction of the blade and handle in order to maintain the blade in a suitable position for test. The screwdriver or screw starter may be held from endwise movement by a stop placed at handle end, provided the stop does not exert pressure on the handle end when assembled and ready for testing.

TABLE XV. Test blocks for torsion test

Nominal tip thickness	Test slot width* (W)	Test slot depth* (D)	Test block designation
Inch	Inch	Inch	
0.015	0.025	0.023	A
0.020	0.030	0.028	B
0.025	0.035	0.033	C
0.030	0.042	0.038	D
0.032	0.044	0.040	E
0.037	0.050	0.045	F
0.042	0.056	0.050	G
0.050	0.064	0.058	H
0.058	0.074	0.066	I
0.070	0.086	0.078	J



* Tolerance: ± 0.001 inch.



Method of torsion test
D x W = inch pounds
D = distance in inches
W = weight in pounds
C is a counterweight used to balance beam

Note: The above method is not intended to restrict the manner in which the required tests shall be made.

Figure 16. Torsional moment test.

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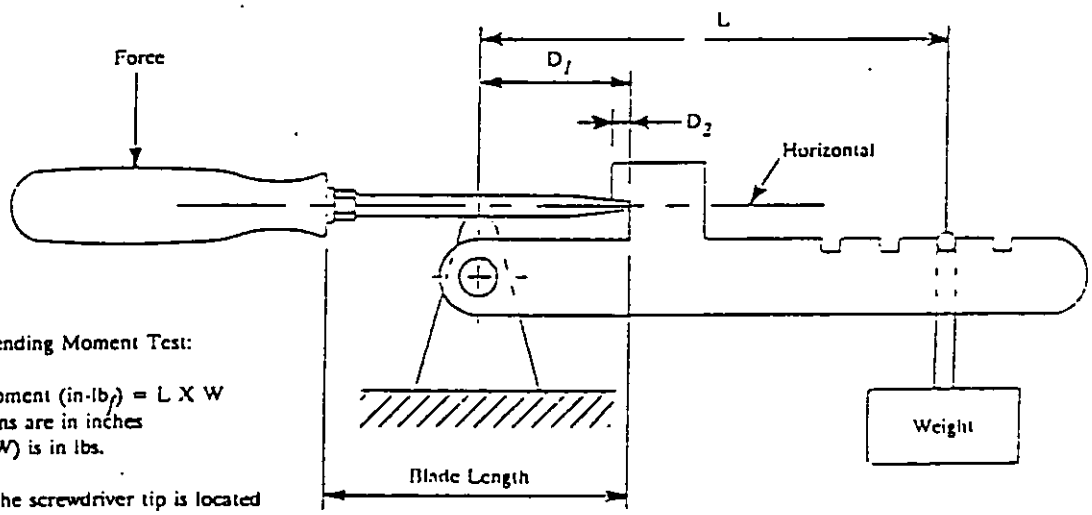
4.4.2.1 Phillips and Reed & Prince cross tip screwdrivers and bits. In lieu of slots when testing these screwdriver, the blade end of each sample under test shall be placed in a die having in its center the corresponding size of Phillips or Reed and Prince recess. The recesses shall be of such depth that the wing spread at the surface shall be as follows:

Phillips point size (number)	Wing spread (inch)
0	0.090
1	0.142
2	0.233
3	0.386
4	0.486

Reed and Prince blade size (inch)	Wing spread (inch)
0.187	0.187
0.250	0.250
0.312	0.312
0.375	0.375

4.4.2.2 Torsional moment. When tested as specified in 4.4.2 through 4.4.2.1, the screwdrivers or screw starters shall not fail nor shown visual indications of weakness of any nature, such as turning of the blade in ferrule or handle, splitting or breaking of the handle, slipping of the bits, twisting, distortion, or damage of any part of the tool or operating mechanism.

4.4.3 Bending moment test. The bending moment test shall be conducted in a manner similar to that shown on figure 17. In this test, the force shall be applied near the middle of the natural grip of the handle, the force to act substantially at right angles to the axis of the screwdriver to lift W. A spring-balance or other suitable load-measuring device may be used in lieu of a dead weight incident to applying the inch-pounds bending test. When tested in this manner, the blades shall not fail, show any visual permanent set, or loosened in the handle.



Method for Bending Moment Test:

- Bending Moment (in-lb$_f$) = L X W
  Dimensions are in inches
  Weight (W) is in lbs.
- A stop for the screwdriver tip is located at a distance D_1, which is equal to 1/2 the blade length, from the bending fulcrum.
- D_2 = 4 X Tip Thickness at Tip.

Note: The above method is not intended to restrict the manner in which the required tests shall be made.

Figure 17. Bending moment test.

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4.4.4 Blade-end toughness test. The flat tip end of the blade shall be inserted in a vise or milled slot to a depth equal to four times the tip thickness of the blade, and the handle pulled slowly as a lever until either a definite permanent bend or a fracture at the blade end results. If fracture occurs, the pieces shall be refitted and the blade end shall show that a definite noticeable permanent bend had occurred prior to the fracture. If no permanent bend of a fractured blade end is apparent, the blade end is brittle and the screwdriver shall be considered to have failed the toughness test requirements.

4.4.5 Handle solvent resistance test. All screwdriver handles and handle cushion grips shall undergo the following test without damage. Handles are to be fully immersed in the test liquids specified (use new handle for each liquid) for 15 minutes at room temperature, removed and allowed to stand for 24 hours: DOT3 brake fluid, gasoline, ethylene glycol, ethyl alcohol, and toluol. There shall be no permanent swelling, surface attack (except for brand or paint removal), or degradation of the applicable performance requirements. The original hardness of the cushion handles shall be Shore A60 to 75 when tested in accordance with ASTM D 2240, and the subsequent hardness after testing shall not be greater than Shore A80.

4.4.6 Impact tests.

4.4.6.1 Impact test at room temperature. The blade end of each applicable screwdriver shall be mounted vertically in a vise affixed to the base of a suitable falling weight impact device. The blade tip shall rest on a solid surface to insure that the blade does not move vertically in the vise. The weight shall be 15 pounds and shall be dropped unrestricted with some means to insure that the full force of the falling weight will be acting normal to the striking surface. The screwdriver handle shall not break, crack, or significantly distort, nor shall the blade penetrate more than 0.62 inches (0.75 inch penetrations acceptable for blade diameters 0.19 inches and smaller) into the handle when the weight has been dropped ten times from the applicable height shown in table XVI. The difference is to be measured after the first and after the tenth drop. The first drop insures that the blade is seated in the handle. Significantly distort (for the purpose of this test) means an increase of at least 5% in the handle diameter, either as a uniform or irregular bulge. In conducting this test, care shall be taken that the impact energy will not be expended in flexing of the blade or in driving the screwdriver tip into the surface on which it rests. The blade shall be shortened or blunted, if necessary, to insure a proper test.

TABLE XVI. Height of drop of 15 pound weight for impact test

Blade diameter (stock size)	Height of drop of 15 pound weight for impact tests
	At room temperature
Inch	Inches
1/8	1-1/2
9/64	1-1/2
5/32	4
3/16	6
7/32	8
1/4	10
9/32	12
5/16	15
11/32	17
3/8 and over	20

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4.4.6.2 Close quarter screwdrivers. For close quarter screwdrivers only, the height of drop shall be decreased to one-half the distance specified in table XVI and the 5/8 inch maximum penetration into the handle shall be reduced to 7/16 inch.

4.4.7 Screw gripping test for type I, class 6, type III, and type VI, class 3 screwdrivers and screw starters.

4.4.7.1 Gripping test. The screw starters shall be connected, by engaging the gripping mechanism to a machine screw of a size for which the point thickness of the blade is suited. The screw shall be driven at least three full turns into a nut of proper thread size. The sample shall be rejected if the gripping mechanism becomes disengaged in the process of starting or driving the screw.

4.4.7.2 Gripper mechanism lifting test. The screw starter shall be connected to a stock screw of proper size for the screw starter point by means of the gripping mechanism. The sample shall be rejected if the screw becomes disengaged from the screw starter in lifting a weight of one and one-half pounds, except for the miniature, type III, which shall be required to lift a weight of 3/4 pound.

4.4.8 Female drive insert. The screwdriver with a female drive insert shall be mounted on a corresponding male drive, and shall retain the male drive with a retention force of no less than four pounds, and shall also withstand a torque load equal to the torque load for the screwdriver without slipping, turning, or breaking.

4.4.9 Surface finish test. Each of the sample screwdrivers shall be tested for surface finish in accordance with ANSI B 46.1 to determine compliance with the surface finish requirements specified (see 3.4.1).

4.4.10 Dielectric strength. When applicable, two sample units shall be tested for dielectric strength. One sample shall be tested after immersion in transformer oil for a period of 5 hours, and one sample shall be tested after immersion in water for a period of 5 hours. After the insulation has be wiped dry, it shall be wrapped in metal foil to within 7/8 inch of the nearest exposed metal part. An electrical lead shall be connected to the exposed tip, and another lead to the foil wrapping, and then subjected to a voltage of not less than 2200 volts, 60 hertz, a.c. After a period of one minute, acceptable insulation covering shall show no signs of breakdown.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table XVII. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE XVII. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style, as applicable (see 1.2).
c. Size (see applicable tables).
d. When insulated blades are required (see 3.4).
e. Handles with 3/8 inch female drive (see 3.6).
f. Whether a cushion grip is to be furnished (see 3.7).
g. Whether additional blades, or blades only, shall be furnished for type I, class 7 screwdrivers (see 3.8.3).
h. When Reed and Prince type bits are to be furnished in lieu of the flat and Phillips tip bits (see 3.12.1).
i. When Reed and Prince type bits are to be furnished separately (see 3.12.1).
j. Size bit required (see 3.12.2).
k. Level of preservation, packaging, and packing required (see 5.1).

6.2 Supersession data. Types and classes of screwdrivers covered in this specification corresponding to those covered in Federal Specification GGG-S-121E are as follows:

GGG-S-121E	GGG-S-121F	Superseded by:
Type I - Flat tip	Type I - Flat tip	
Class 1 - Cabinet, Straight sides	Deleted	B107.15, Class 1
Design A - Plain	Deleted	B107.15, Type 1, Class 1, Design A
Class 3 - Close quarter	Deleted	B107.15, Type 1, Class 2
Class 4 - Heavy duty, flared sides	Class 4 - Heavy duty, flared sides	
Style 2 - Steel handle with wood inserts	Style 2 - Steel handle with wood inserts	
Class 5 - General purpose, flared sides	Deleted	B107.15, Type 1, Class 2
Design A - Plain, medium heavy duty	Deleted	B107.15, Type 1, Class 2, Design A
Design B - Bolster, forged, with hexagon-wrench grip	Deleted	B107.15, Type 1, Class 2, Design B
Class 6 - Flared sides with screw starter, external gripping	Class 6 - Flared sides with screw starter, external gripping	
Class 7 - Pocket, removable blade	Class 7 - Pocket, removable blade	
Class 8 - Pocket, clip	Deleted	B107.15, Type 1, Class 1, Design D
Style 1 - Magnetized tip	Deleted	B107.15, Type 1, Class 1, Design D
Style 2 - Nonmagnetized tip	Deleted	B107.15, Type 1, Class 1, Design D
Type III - Screw starters, hand	Type III - Screw starters, hand	
Type IV - Offset	Type IV - Offset	
Class 1 - Double offset, single flat tip each end	Class 1 - Double offset, single flat tip each end	
Class 2 - Four-way offset, double flat tip each end	Class 2 - Four-way offset, double flat tip each end	
Style 1 - Double offset	Style 1 - Double offset	
Style 2 - Single offset	Style 2 - Single offset	
Type V - Ratchet	Type V - Ratchet	
Class 1 - Common, straight	Class 1 - Common, straight	
Class 2 - Double spiral, automatic return	Deleted	Commercial technical description
Style 1 - Light duty	Deleted	Commercial technical description
Style 2 - Medium duty	Deleted	Commercial technical description
Style 3 - Heavy duty	Deleted	Commercial technical description

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GGG-S-121E	GGG-S-121F	Superseded by:
Type VI - Cross tip, straight	Type VI - Cross tip, straight	
Class 1 - Phillips	Deleted	B107.15, Type II
Class 2 - Reed and Prince	Class 2 - Reed and Prince	
Class 3 - Phillips, with screw starter, external gripping, with cushion grip	Class 3 - Phillips, with screw starter, external gripping, with cushion grip	
Type VII - Screwdriver, spark testing	Type VII - Screwdriver, spark testing	
Type VIII - Bit set and handle	Deleted	Commercial technical description
Class 1 - Plastic handle with clutch drive	Deleted	Commercial technical description
Class 2 - Bit, Phillips, cross tip	Deleted	Commercial technical description
Class 3 - Bit, cabinet type tip	Deleted	Commercial technical description
Type IX - Flat tip - cross tip, close quarter	Deleted	Commercial technical description
Type X - Clutch tip	Deleted	Commercial technical description

MILITARY INTERESTS:

DoD has waived coordination on revision and amendments to this Federal Specification until further notice.

PREPARING ACTIVITY:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
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